

TriVoKo

Phase 10 - Smart search

Time	Start date	Finish date	Status
1 week			Not started

Goal: Standout #4: search that forgives typos, suggests while you type, and shows filters with counts.

Real-life example: A good shop assistant understands "iphnoe" means iPhone, and says "we have 12 Samsung and 8 Apple phones" before you ask.

Step checklist

★ = you type the key logic (Claude explains first, then reviews). "You" = only you can do it.

#	Step	Who	Done
1	Spike: Hibernate Search 8 with the Lucene backend on Spring Boot 4; index folder; rebuild on start	Claude	☐
2	Map products: full-text fields (title, brand, description), keyword fields for facets	Claude	☐
3	Search API: fuzzy match, filters, facets with counts, sort	Claude	☐
4	Suggest API: suggestions after 2 letters	Claude	☐
5	Frontend: search box with suggestions (debounce, keyboard), result page with facet counts	Claude	☐
6	"Customers also bought" (SQL, cached 1 hour)	Claude	☐
7	Decision: keep Lucene or fall back to MySQL FULLTEXT for hosting - written in docs/decisions.md	Together	☐

Output of this phase

- Smart search with typo tolerance, suggestions and facets

Done when (check before moving on)

Check	Passed
"iphnoe" finds iPhone	☐
Suggestions appear while typing	☐

Explain it back (in your own words)

Question	Done
What is an index, and why is search not a SQL LIKE query?	☐

Question	Done
How does fuzzy matching find "iphnoe"?	☐

Finish routine

Check	Done
All tests pass and CI is green	☐
Committed and pushed	☐
Done-when checks shown working in the browser	☐
Session summary + SESSION_LOG written	☐

Skills you practise: Full-text search, Lucene, analyzers, facets, debounce

Where your work goes: `backend\...\search`, search components in `frontend`

My notes
